



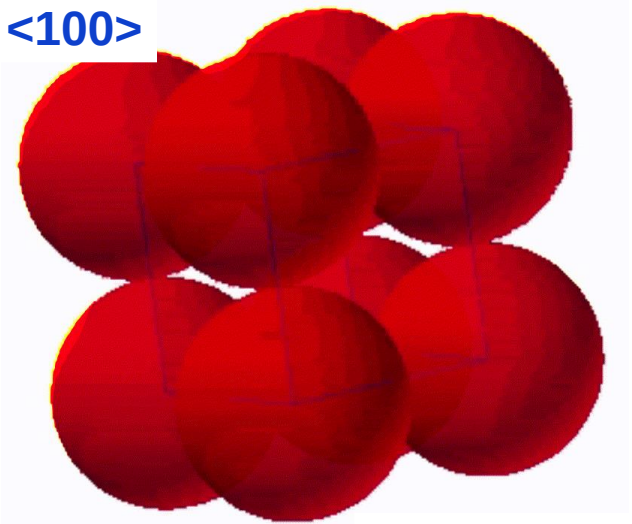
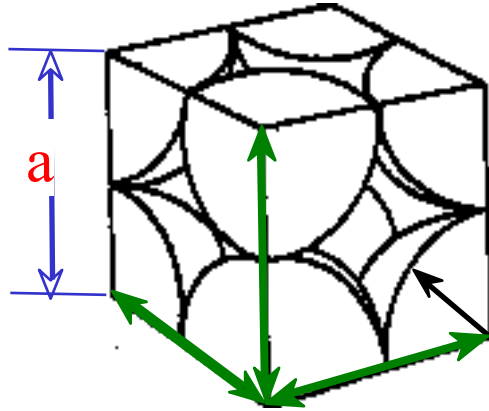
## Outline:

- Review of Crystal structure
- Stacking Sequence
- Theoretical Density
- Crystallographic Directions and Planes
  - *examples*



# Review: *Crystal Structure*

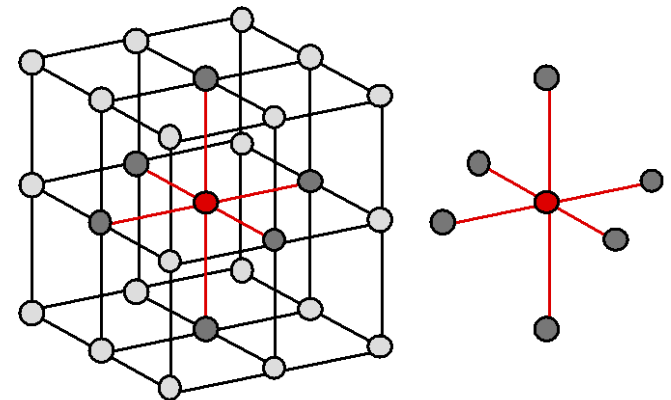
- Close-packed directions are ..... **cube edges  $\langle 100 \rangle$**



This is a ... **Simple Cubic**

- Contains = .... **1 atom** .....
- Coordination # = 6** (*nearest neighbors*)

$$\text{APF} = \frac{1 \frac{4}{3} \pi (0.5a)^3}{a^3}$$



- APF for a simple cubic structure = ..... **0.52 (52%)**



# Review: Crystal Structure

- Close packed directions are .....

- Coordination # = . **8**

$$\text{APF} = \frac{2 \cdot \frac{4}{3} \pi \left( \frac{\sqrt{3}a}{4} \right)^3}{a^3}$$

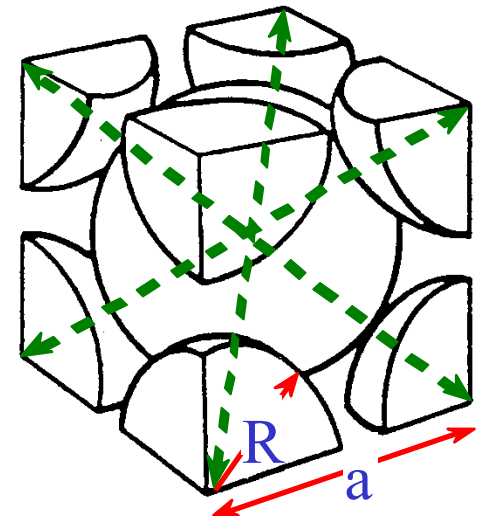
- APF for a BCC structure = ..... **0.68 (68%)**

body diagonals  $\langle 111 \rangle$



**Note:** All atoms are identical; the center atom is shaded differently only for ease of viewing.

This is a .... **BCC** cell

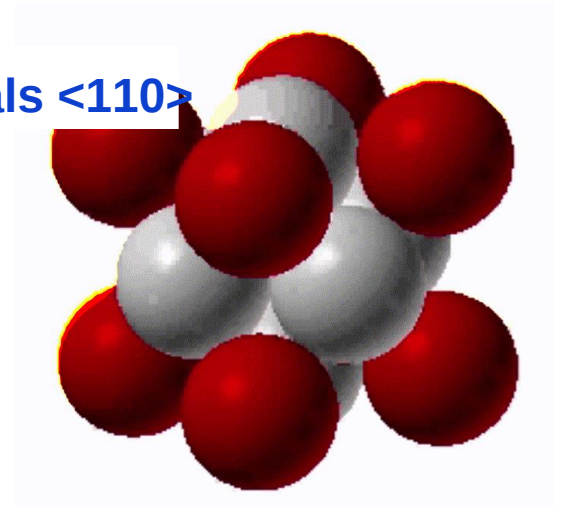




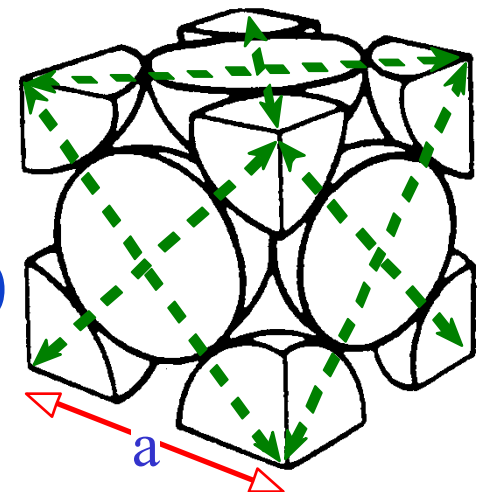
# Review: Crystal Structure

- Close packed directions are ..... **face diagonals  $\langle 110 \rangle$**
- Coordination # = ..... **12**

$$\text{APF} = \frac{4 \cdot \frac{4}{3} \pi \left( \frac{\sqrt{2}a}{4} \right)^3}{a^3}$$



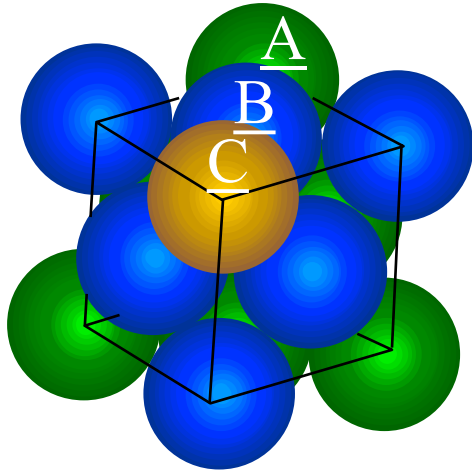
This is a .. **FCC** it cell



- APF for a FCC structure = ..... **0.74 (74%)**



# FCC Stacking Sequence

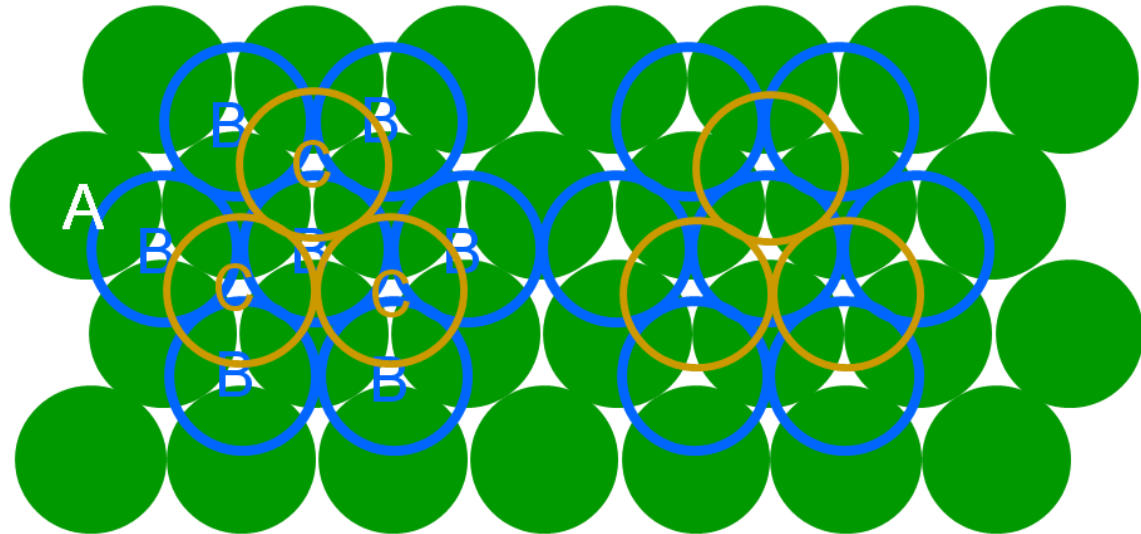


- FCC Unit Cell

A sites

B sites

C sites



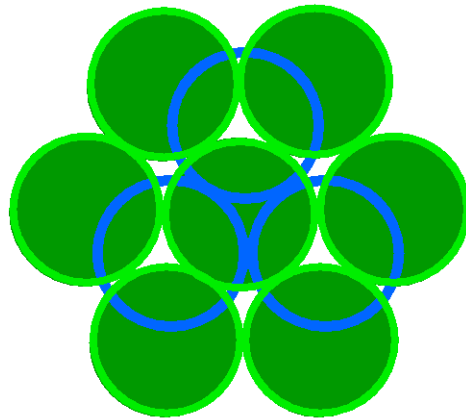
2D Projection

- ABCABC...etc Stacking Sequence



# Review: *Crystal Structure*

- Coordination # = **12**
- APF = **0.74 (74%)**



*2D Projection*

**Top layer**

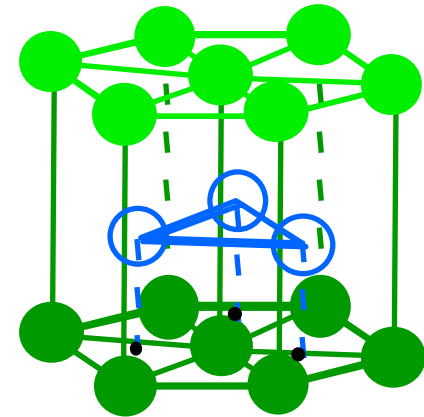
**Middle layer**

**Bottom layer**

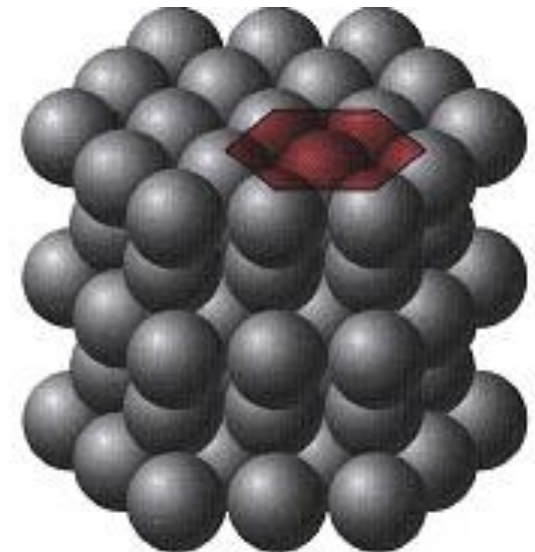
A sites

B sites

A sites



This is a .. **HCP** t cell



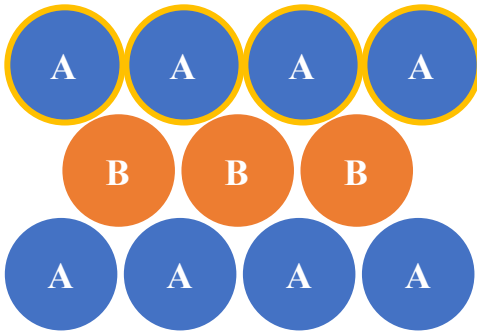
- Stacking Sequence: **ABABAB...**



# HCP vs. FCC: *the Third Layer Decides*

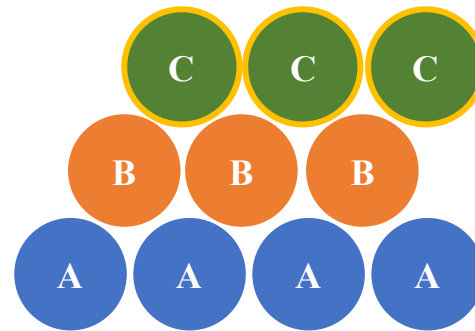
*Both start the same way: a close-packed A layer, then a B layer in the hollows.*

## **HCP — ABAB**



Third layer sits directly above the A atoms. The C hollows stay empty, so the pattern repeats every two layers: ABAB...

## **FCC — ABCABC**



Third layer shifts into the empty C hollows — a new position. Only the fourth layer returns to A: ABCABC...

Same packing, different order:  $CN = 12$  and  $APF = 0.74$  for both. The stacking sequence is what differs — and it is why FCC and HCP deform so differently.



# Example Problem

If you know

- the crystal structure,
- the atomic radius
- the atomic weight,

*you can calculate the density of a particular material*

## Example:

Copper has an atomic radius 0.128 nm an FCC crystal structure and an atomic weight of 63.5 g/mol. Calculate its density.



# Crystallographic Directions, and Planes

Now that we know how atoms arrange themselves to form crystals, *we need a way to identify directions and planes of atoms.*

## Why?

- ✓ Deformation under loading ( **slip** ) occurs on certain crystalline planes and in certain crystallographic directions.
- ✓ Before we can predict how materials fail, we need to know what modes of failure are more likely to occur.
- ✓ Other properties of materials (*electrical conductivity, thermal conductivity, elastic modulus*) can vary in a crystal with orientation.

Directions and planes are described using three integers -

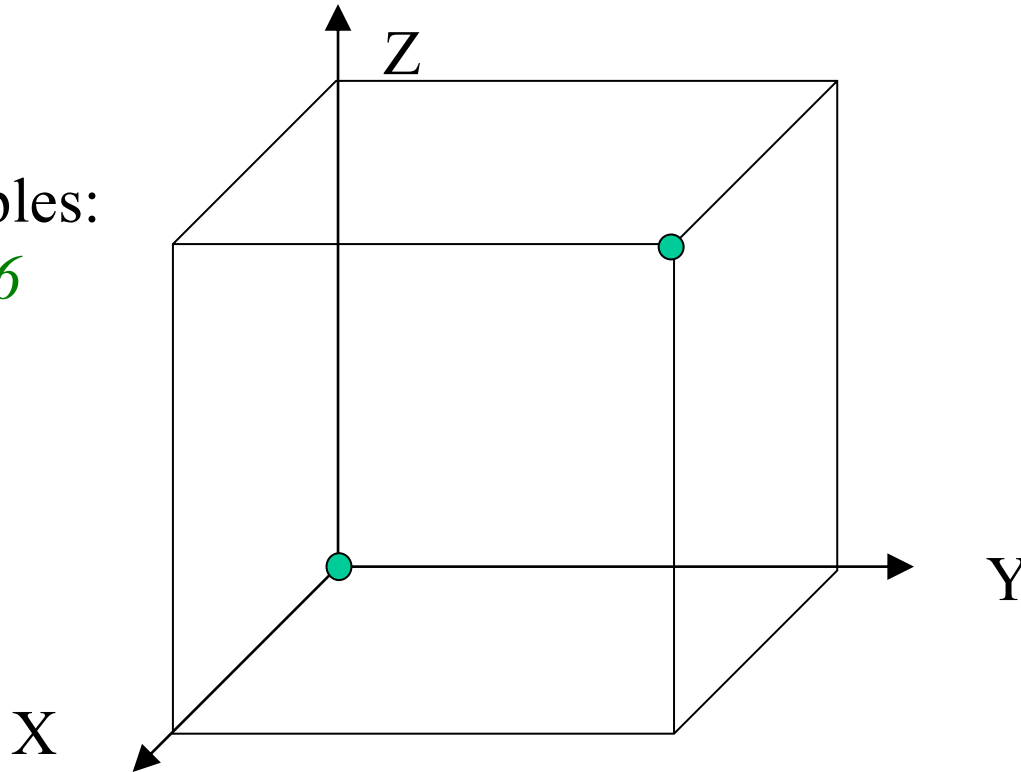
**Miller** **Indices**



# Point Coordinates

Point position specified in terms of its coordinates as fractional multiples of the unit cell edge lengths

See examples:  
*3.5 and 3.6*





# General Rules for Lattice Directions, Planes & Miller Indices

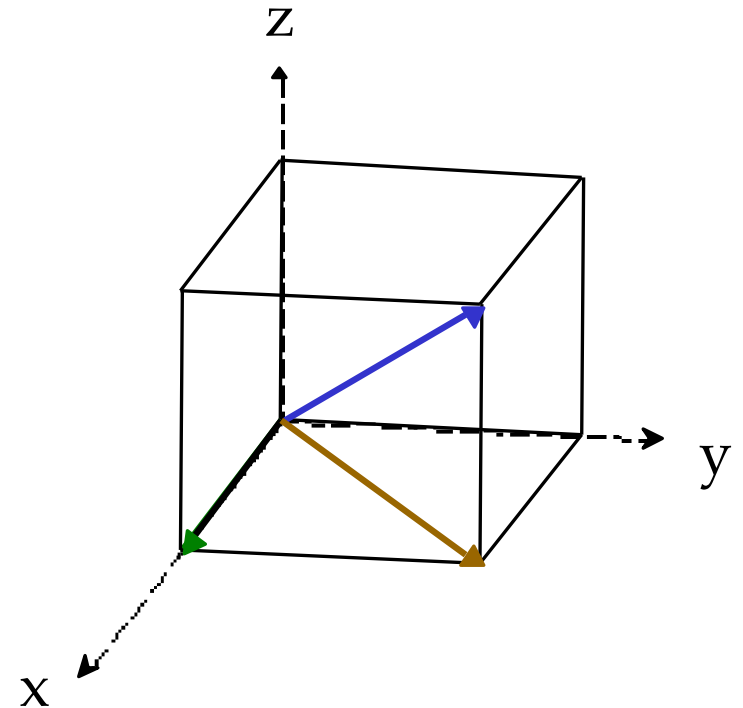
- Miller indices used to express lattice planes and directions
  - x, y, z are the axes (on arbitrarily positioned origin)
    - *in some crystal systems these are not mutually  $\perp$*
  - a, b, c are lattice parameters (*length of unit cell along a, b, c sides*)
  - h, k, l are the Miller indices for planes and directions - expressed as planes: (hkl) and directions: [hkl]
  - Conventions for naming
    - There are **NO COMMAS** between numbers
    - Negative values are expressed with a bar over the number
      - *Example: -2 is expressed  $\bar{2}$*
- Crystallographic direction:
    - [123]
    - [100]
    - ... etc.



# Miller Indices for Directions

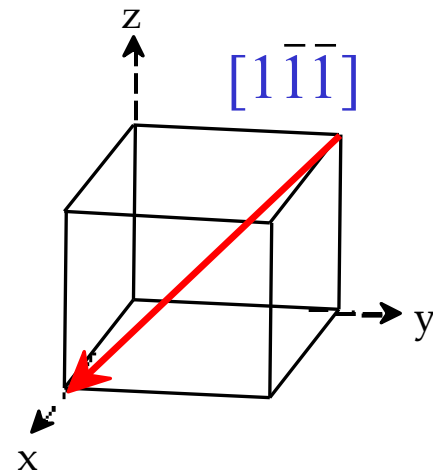
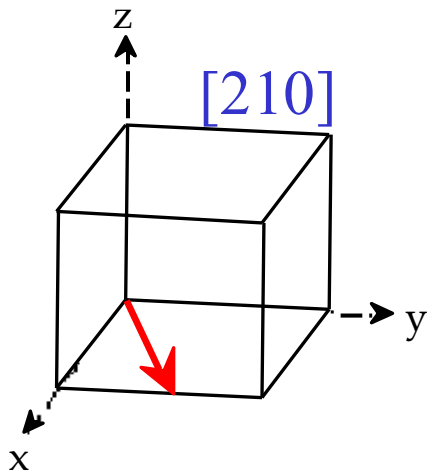
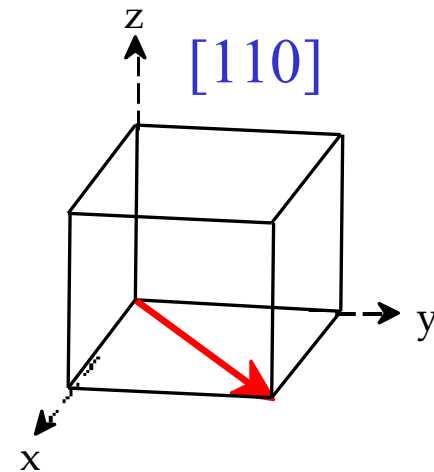
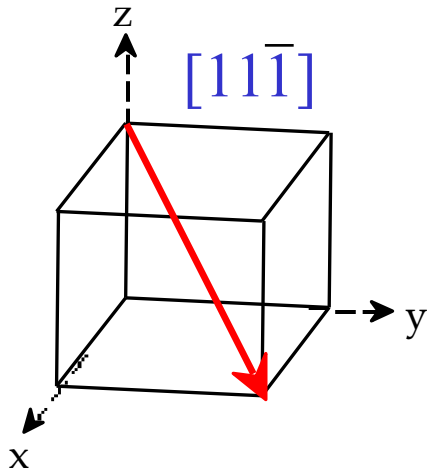
## Method

- Draw vector, define tail as origin.
- Determine length of the vector **projection** in unit cell dimensions,  $a$ ,  $b$ , and  $c$
- Remove fractions by multiplying by smallest possible factor
- Enclose in square brackets



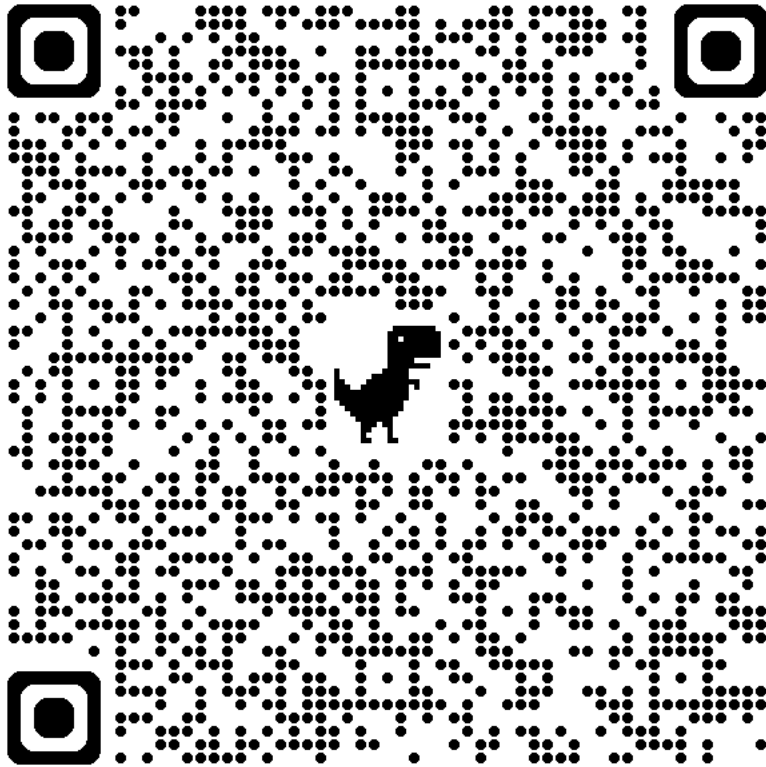


# Example - Naming Directions



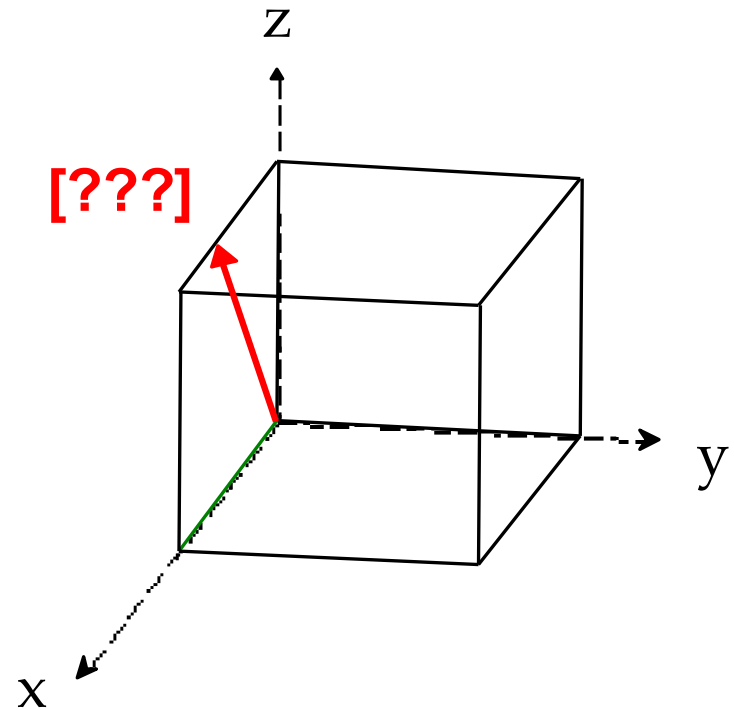


# Miller Indices for Directions



Use the QR code shown in the classroom. This one won't work.

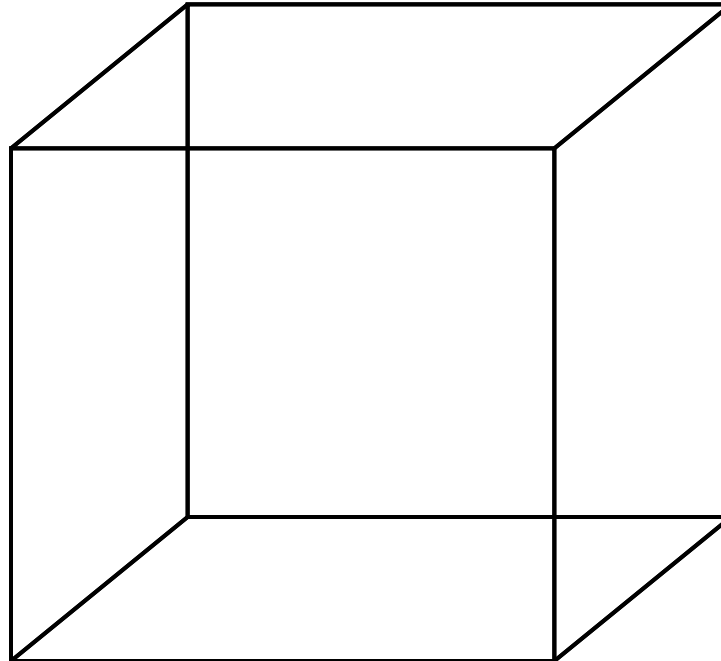
What is  $[? ? ?]$





# Example - Drawing Directions

- Draw  $[11\bar{2}]$  and  $[\bar{1}\bar{1}\bar{1}]$

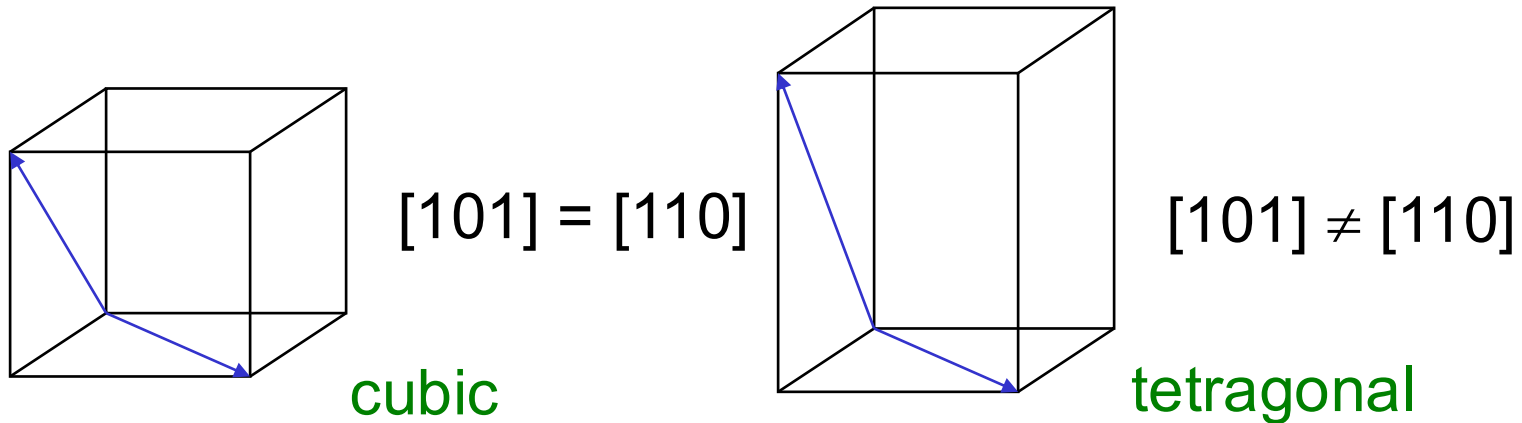


*The HW problems will give you more practice with directions*



# Families of Directions

- Equivalence of directions

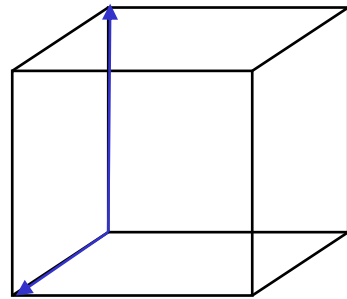


- $\langle 123 \rangle$  Family of directions
  - $[123], [213], [312], [132], [231], [321]$   
– *only in a cubic crystal*

*In the **cubic system** directions having the same indices regardless of order or sign are *equivalent*.*



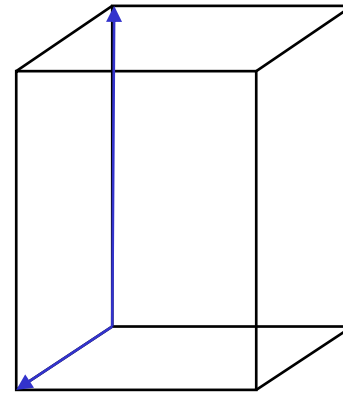
# Isotropy vs. Anisotropy in Single Crystals



$$[100] = [001]$$

cubic

Isotropic



$$[100] \neq [001]$$

tetragonal

Anisotropic

- Properties are independent of direction → isotropic material
- Directionality of properties → anisotropic material



# Miller Indices for Planes

- $(hkl)$  Crystallographic plane
- $\{hkl\}$  Family of crystallographic planes
  - e.g.  $(hkl)$ ,  $(lhk)$ ,  $(hkl)$  ... etc.

*In the cubic system planes having the same indices regardless of order or sign are equivalent*

- **Hexagonal** crystals can be expressed in a four index system  $(u\ v\ t\ w)$ 
  - *Can be converted to a three index system using formulas*

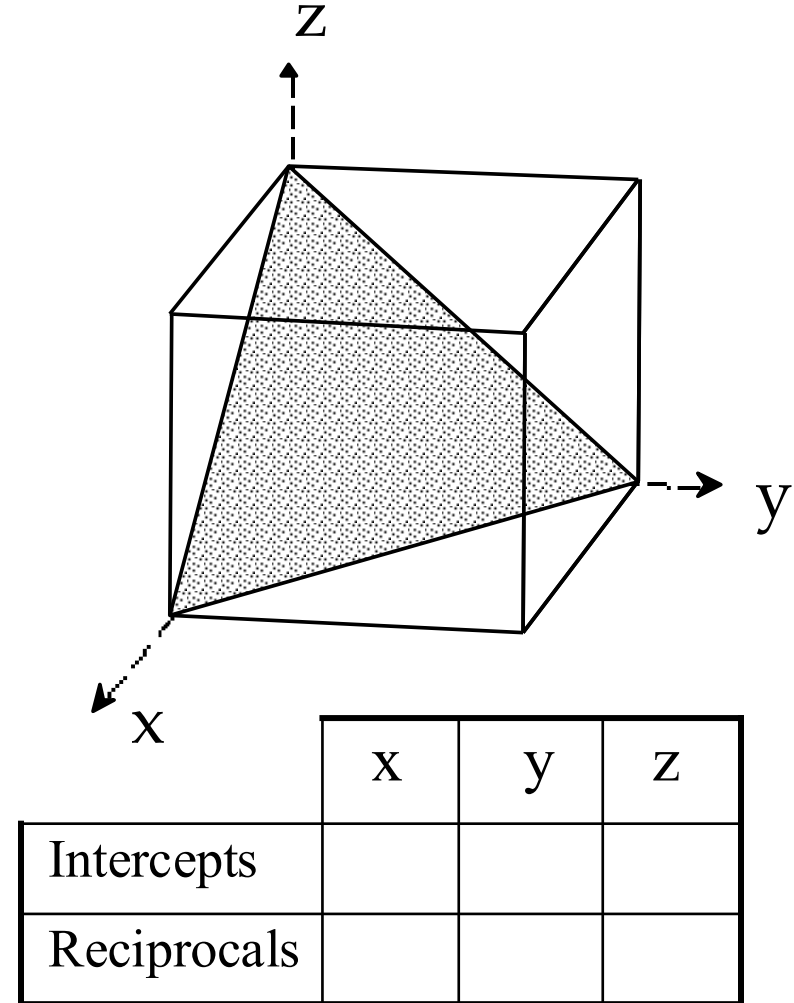


# Miller Indices for *PLANES*

## *Method*

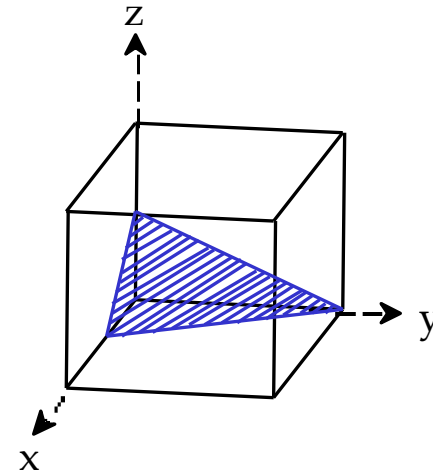
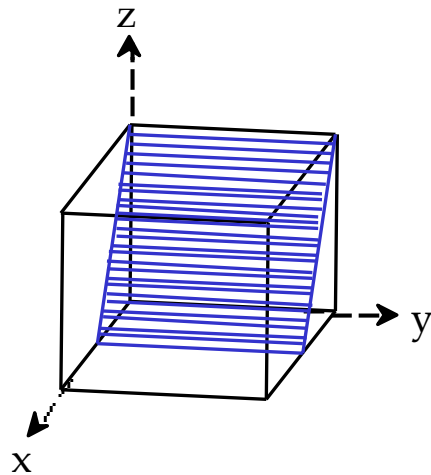
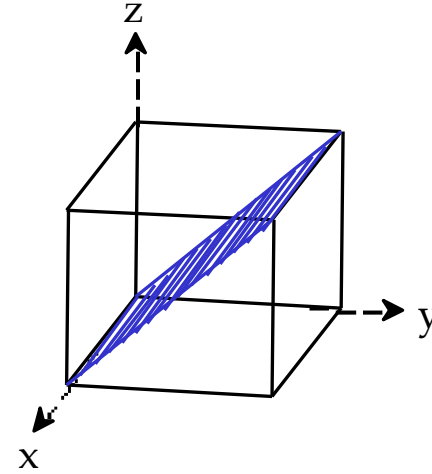
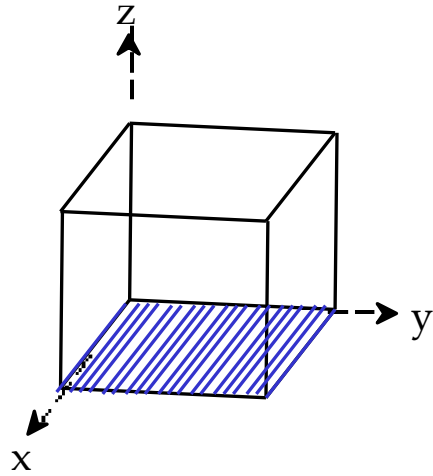
- If the plane passes through the origin, select an equivalent plane or move the origin
- Determine the intersection of the plane with the axes in terms of a, b, and c
- Take the reciprocal ( $1/\infty = 0$ )
- Convert to smallest integers (*optional*)
- Enclose by parentheses

*see example 3.10*





# Crystallographic Planes





Next time:

# Linear and Planner Densities